

Contents

Validation methodologies for basis function expansion (BFE) potentials: a literature review

Summary table

Discussion by validation aspect

Most relevant precedents for your specific test

Full references

Notes on scope and confidence

1

1

10

11

11

12

Validation methodologies for basis function expansion (BFE) potentials: a literature review

Scope: refereed papers in MNRAS, ApJ (incl. ApJL/ApJS), A&A, AJ, and comparable-tier venues that present a concrete methodology for testing how well a basis function expansion (BFE) / self-consistent field (SCF) representation reproduces the true gravitational potential, density, or dynamics of an N-body or cosmological system. Organized by the specific aspect of the approximation each methodology targets, since different papers stress-test different failure modes (spatial truncation, temporal under-sampling, coefficient noise, force accuracy, downstream orbit/stream fidelity).

Summary table

#	Reference	Validation aspect	Method	Headline result
1	Hernquist & Ostriker 1992, ApJ, 386, 375	Spatial truncation / basis choice (foundational)	Compares SCF potential re-constructed from a finite (n,l,m) truncation against the input analytic/N-body density; basis chosen so the lowest-order term matches the target profile	Establishes that a basis whose zeroth-order term already resembles the target profile converges with far fewer terms than a generic (e.g. spherical Bessel) basis

#	Reference	Validation aspect	Method	Headline result
2	Lowing, Jenkins, Eke & Frenk 2011, MNRAS, 416, 2697	Spatial order and temporal cadence	Fits Hernquist-Ostriker BFE to a cosmological halo at each of ~ 400 snapshots (~ 25 Myr apart), linearly interpolates coefficients between snapshots, integrates test-particle/subhalo orbits and compares to orbits in the raw N-body potential	12 radial $\times$ 6 angular terms reproduce the potential to a few percent; linear interpolation across ~ 25 Myr snapshots gives orbits that track the true (higher-cadence) potential well, but degrades for orbits with short pericentric periods
3	Lilley, Sanders, Evans & Erkal 2018, MNRAS, 476, 2092	Basis-set choice / convergence rate	Derives a new biorthogonal potential-density basis from the Hankel transform of Laguerre polynomials, tuned to the NFW asymptotic falloff; compares the power-law decay rate of expansion coefficients (C_{n0}) against the classic Hernquist-Ostriker basis for the same target halo	New basis coefficients decay as n^{-2} vs. n^{-3} for Hernquist-Ostriker on an NFW-like target, i.e. fewer terms needed for a basis matched to the target's asymptotic slope

#	Reference	Validation aspect	Method	Headline result
4	Lilley, Sanders & Evans 2018, MNRAS, 478, 1281	Basis-set choice / generality	Extends the biorthogonal family to a two-parameter double-power-law basis and tests reconstruction accuracy across a range of inner/outer density slopes	Demonstrates which basis-family parameters best match a given halo's inner/outer slope, directly informing basis choice for non-Hernquist/NFW targets
5	Wang, Athanassoula & Mao 2020, A&A, 639, A38	Basis geometry (spherical harmonic vs. cylindrical)	Directly compares the Hernquist-Ostriker spherical-harmonic expansion against the Vasiliev & Athanassoula cylindrical (CylSP) expansion for a family of generalized Dehnen density profiles of varying flattening	CylSP is more accurate than the spherical-harmonic expansion for nearly all inner slopes and flattenings tested — basis geometry should match the target's intrinsic symmetry (relevant if your satellite or host departs from spherical)

#	Reference	Validation aspect	Method	Headline result
6	Vasiliev 2019, MNRAS, 482, 1525 (the AGAMA paper)	Spatial/radial-grid resolution; smooth vs. discrete potential	Builds multipole and “CylSpline” potential approximations directly from N-body snapshots and from analytic density laws; compares the resulting smooth potential to both the exact analytic potential and to the raw (noisy) potential implied by the N-body particles themselves	With roughly 15–30 radial grid points the smoothed expansion is <i>more</i> accurate than evaluating the potential directly from the discrete N-body particles, because the expansion suppresses Poisson (shot) noise; accuracy degrades sharply below ~15 points

#	Reference	Validation aspect	Method	Headline result
7	Sanders, Lilley, Vasiliev, Evans & Erkal 2020, MNRAS, 499, 4793 (“Models of distorted and evolving dark matter haloes”)	Spatial order and temporal cadence, jointly, for a cosmological zoom-in halo	Fits BFE (both Hernquist-Ostriker-type and spline-radial bases) to every snapshot of a Milky Way-like zoom-in simulation, explicitly varies both expansion order and snapshot spacing, and measures the resulting error in reconstructed satellite/test-particle orbits (position, energy, angular momentum) relative to orbits integrated directly in the simulation	≥ 15 radial and ≥ 6 angular terms needed; a conventional Hernquist-Ostriker expansion is hard to beat despite testing alternative bases; time evolution over 5 Gyr introduces $\sim 15\%$ uncertainty in recovered satellite orbital parameters, comparable to observational/potential-model uncertainties — this paper is the closest direct precedent for your cadence experiment

#	Reference	Validation aspect	Method	Headline result
8	Garavito-Camargo et al. 2019, ApJ, 884, 51 ("Hunting for the Dark Matter Wake...")	Downstream astrophysical fidelity (density/kinematic wake structure)	Compares the dark-matter wake structure (density and velocity perturbations) induced by an LMC-mass satellite as measured directly in a live N-body simulation vs. as reconstructed from a BFE representation of the same simulation	BFE reproduces both the local (near-satellite) and global (halo-wide) wake morphology seen in the raw simulation, validating BFE as a tool for extracting perturbation structure, not just bulk potential shape

#	Reference	Validation aspect	Method	Headline result
9	Garavito-Camargo et al. 2021, ApJ, 919, 109	Temporal cadence for a <i>live, responding</i> host (coupled MW+LMC wake)	Fits time-dependent BFE coefficients to a live, self-consistently responding MW halo perturbed by an infalling LMC-mass satellite across the simulation's snapshot cadence, and evaluates how well the interpolated, time-varying BFE reproduces the halo's asymmetric, evolving structure at arbitrary intermediate times	Demonstrates BFE can track a genuinely time-evolving (not just orbiting-test-particle) host response, directly relevant if you want to eventually let the host itself respond rather than stay static

#	Reference	Validation aspect	Method	Headline result
10	Petersen, Weinberg & Katz 2022, MNRAS, 510, 6201 (the EXP code paper)	Force accuracy vs. direct summation; coefficient noise / adaptive basis selection	Runs matched N-body integrations using (a) BFE-derived forces (adaptive, empirically-optimized basis) and (b) direct-summation/tree forces on the same initial conditions; compares per-particle force errors and energy conservation; also describes signal-to-noise-based truncation of noisy high-order coefficients	Median force error of 0.52% for the BFE vs. 1.02% for direct summation on the same test problem — BFE forces can be <i>more</i> accurate than direct summation because the expansion suppresses discreteness noise, provided the basis is adapted to the mass distribution

#	Reference	Validation aspect	Method	Headline result
11	Arora, Sanderson, Regan, Garavito- Camargo, Bregou, Panithanpaisal, Wetzel, Cunningham, Loebman, Dropulic & Shipp 2024, ApJ, 977, 23 ("Efficient and accurate force replay...")	Temporal cadence, directly, in a cosmological-baryonic (FIRE-2) simulation; force- and orbit-level error budgets	Fits low-order BFE (spherical harmonics for the halo, azimuthal harmonics for the disk) to each snapshot of a FIRE-2 Milky Way-mass galaxy at ~25 Myr cadence over 4 Gyr, linearly interpolates forces between snapshots, and separately repeats the experiment at 200 Myr cadence; measures force- reconstruction error against the simulation's true forces and orbit- integration error (position, energy, angular momentum) over multiple orbital periods	At 25 Myr cadence, 95% of particles (outside self-gravitating subhalos) have force errors $\leq 4\%$; most orbits have position errors $\leq 10\%$ for 2–3 orbital periods; after 4 Gyr, 43%/70% of orbits retain energy/angular- momentum errors within 10%; central/high-frequency orbits show the largest errors; integrals of motion remain robust even at 200 Myr cadence even though absolute positions do not — this is the paper methodologically closest to what you're planning, including its explicit framing around satellite disruption

#	Reference	Validation aspect	Method	Headline result
12	Petersen, Roule, Fouvry, Pichon & Tep 2024, MNRAS, 530, 4378 (LinearResponse.jl)	Cross-validation against independent theory (not N-body)	Predicts the linear dynamical response of self-gravitating spheres/discs (Plummer model, isochrone, Mestel disc) analytically via linear response theory in the same BFE basis used by EXP, then compares to N-body measurements made with the EXP code	Analytic linear-response predictions and EXP’s BFE-based N-body measurements agree well, providing a validation route for BFE-based dynamics that is independent of “compare to a higher-resolution/finer-cadence N-body run”

Discussion by validation aspect

Spatial truncation order and basis-set choice (rows 1, 3, 4, 5, 6, 7). The oldest and most standard check: fix the target density/potential, vary the number of radial and angular terms retained, and measure the residual between the truncated BFE and the true field. Hernquist & Ostriker (1992) established the principle that matching the lowest-order basis function to the target’s overall shape lets you truncate at low order; Lilley et al. (2018a,b) and Wang et al. (2020) show this matters quantitatively — an NFW-like halo needs a different basis (or coordinate system) than a Hernquist profile to converge at the same order, and Vasiliev (2019) shows that spatial/radial resolution choices trade off against noise suppression, not just accuracy. None of this addresses your specific concern (temporal cadence), but it sets the “spatial error floor” against which any temporal error should be compared, since the two error sources are usually reported together in the papers below.

Temporal cadence and coefficient interpolation (rows 2, 7, 9, 11) — the category most directly relevant to your question. All four papers fit BFE coefficients snapshot-by-snapshot and interpolate (typically linearly) between them, then quantify how the resulting time-dependent potential degrades orbit integration relative to the “true” (fine time-resolution or continuously-computed) potential. Lowing et al. (2011) established the ~ 25 Myr cadence as adequate for halo-scale orbits in their cosmological box; Sanders et al. (2020) is the first to jointly vary spatial order *and* cadence and attribute error budget between them; Arora et al. (2024) is the most quantitative and closest analogue to what you’re planning — it explicitly tests two cadences (25 Myr vs. 200 Myr), separates force-level from orbit-level error, and flags that orbits spending more time near the galactic center (i.e., short dynamical times, exactly your worry) show the largest degradation. Garavito-Camargo et al. (2021) is the only one of these that lets the *host* itself respond (rather than treating the host as fixed and only the satellite/test-particles as evolving), which is a natural next step after your static-analytic-host test.

Force/energy accuracy vs. direct summation (rows 6, 10, 11). These papers benchmark BFE-derived forces against a “ground truth” computed by direct summation or a tree code on the same particle distribution, independent of any time-dependence. Petersen, Weinberg & Katz (2022) is notable for showing BFE forces can *beat* direct summation because the expansion filters Poisson noise

— a useful point if you’re ever tempted to treat direct N-body forces as automatically “more true” than the BFE.

Coefficient noise and adaptive basis selection (row 10, discussed within Petersen et al. 2022). Distinguishing physical coefficients from those dominated by finite-N discreteness noise is a prerequisite for trusting high-order terms; EXP’s adaptive/empirical basis construction and signal-to-noise-based coefficient selection is the most complete refereed treatment in your target journal list.

Downstream astrophysical / orbit fidelity (rows 8, 9, 11 again). Beyond generic force/orbit-position error metrics, Garavito-Camargo et al. (2019, 2021) validate that BFE reproduces physically meaningful derived structures (the dark-matter wake), not just pointwise force accuracy — a useful template if you want your validation to report on stream morphology rather than only energy/angular-momentum conservation.

Independent cross-validation against theory (row 12). `LinearResponse.jl` is a genuinely different validation strategy: instead of comparing BFE-based dynamics to a finer-resolution/finer-cadence N-body run (as in rows 2, 7, 9, 11), it compares to an analytic linear-response prediction computed in the same basis. Not directly applicable to your cadence question, but worth knowing about as an alternative validation axis if N-body ground truth is ever unavailable or too expensive.

Most relevant precedents for your specific test

If you want directly comparable prior work before designing your own cadence experiment, prioritize:

- **Arora et al. 2024 (ApJ 977, 23)** — closest in spirit and method to your planned test; explicitly compares two snapshot cadences, separates force- from orbit-level error, and frames the motivation around satellite disruption.
- **Sanders et al. 2020 (MNRAS 499, 4793)** — the most careful joint treatment of spatial order and temporal cadence, with orbit-error quantification for Milky Way satellites.
- **Lowing et al. 2011 (MNRAS 416, 2697)** — the original demonstration that ~ 25 Myr cadence + linear interpolation is adequate for halo-scale (not necessarily central/short-period) orbits.

Full references

1. Hernquist, L., & Ostriker, J. P. 1992, ApJ, 386, 375, “A Self-consistent Field Method for Galactic Dynamics”
2. Lowing, B., Jenkins, A., Eke, V., & Frenk, C. 2011, MNRAS, 416, 2697, “A halo expansion technique for approximating simulated dark matter haloes”
3. Lilley, E. J., Sanders, J. L., Evans, N. W., & Erkal, D. 2018, MNRAS, 476, 2092, “Galaxy halo expansions: a new biorthogonal family of potential-density pairs”
4. Lilley, E. J., Sanders, J. L., & Evans, N. W. 2018, MNRAS, 478, 1281, “A two-parameter family of double-power-law biorthonormal potential-density expansions”
5. Wang, Y., Athanassoula, E., & Mao, S. 2020, A&A, 639, A38, “Basis function expansions for galactic dynamics: Spherical versus cylindrical coordinates”
6. Vasiliev, E. 2019, MNRAS, 482, 1525, “AGAMA: action-based galaxy modelling architecture”
7. Sanders, J. L., Lilley, E. J., Vasiliev, E., Evans, N. W., & Erkal, D. 2020, MNRAS, 499, 4793, “Models of distorted and evolving dark matter haloes”
8. Garavito-Camargo, N., Besla, G., Laporte, C. F. P., et al. 2019, ApJ, 884, 51, “Hunting for the Dark Matter Wake Induced by the Large Magellanic Cloud”
9. Garavito-Camargo, N., Besla, G., Laporte, C. F. P., et al. 2021, ApJ, 919, 109, “Quantifying the Impact of the Large Magellanic Cloud on the Structure of the Milky Way’s Dark Matter Halo Using Basis Function Expansions”
10. Petersen, M. S., Weinberg, M. D., & Katz, N. 2022, MNRAS, 510, 6201, “EXP: N-body integration using basis function expansions”
11. Arora, A., Sanderson, R. E., Regan, C., Garavito-Camargo, N., Bregou, E., Panithanpaisal, N., Wetzel, A., Cunningham, E. C., Loebman, S. R., Dropulic, A., & Shipp, N. 2024, ApJ, 977, 23, “Efficient and Accurate Force Replay in Cosmological-baryonic Simulations”
12. Petersen, M. S., Roule, M., Fouvry, J.-B., Pichon, C., & Tep, K. 2024, MNRAS, 530, 4378, “Predicting the linear response of self-gravitating stellar spheres and discs with `LinearResponse.jl`”

Notes on scope and confidence

- All 12 references were confirmed via their journal landing pages (Oxford Academic for MNRAS, IOPscience for ApJ, A&A's own site) or NASA ADS during this search, not from memory alone, and all fall within MNRAS/ApJ/A&A as requested.
- I deliberately excluded a widely-referenced but harder-to-pin-down Weinberg paper on coefficient signal-to-noise regularization/truncation, since I could not verify its exact title/venue with confidence in this search session; the same coefficient-selection methodology is, however, documented and properly citable via Petersen, Weinberg & Katz (2022) (row 10), which is fully verified.
- I did not include preprints without a confirmed refereed venue (e.g., very recent 2025/2026 arXiv-only papers on BFE applications) to stay strictly within your journal-quality requirement; let me know if you want those added with an explicit “preprint, not yet refereed” f